CompoundIQ Phase Summary

Portfolio-friendly overview of the group capstone project and my individual contribution

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Group Project: CompoundIQ

What this presentation shows

- A clearer summary of what happened in Phases 1–5 of the CompoundIQ capstone.
- A transparent explanation of which parts were group work and which parts I personally completed.
- The skills I built in drug data engineering, feature engineering, project communication, and GitHub organization.

My individual role in the project

- DrugBank data pipeline and feature engineering work.
- Created and presented the Phase 2 Data Pipeline & EDA presentation.
- Reviewed the broader EDA notebook to understand how other datasets connected to the final pipeline.
- Organized and cleaned up the group GitHub repository for submission.

Project Overview and Team Context

CompoundIQ was a group capstone project focused on safe 3-drug combination discovery.

P 1 — P 2 — P 3 — P 4 — P 5

What CompoundIQ aimed to do

- ➢ Take a therapeutic mechanism written in plain English.
- ➢ Use AI components to generate candidate compounds and evaluate drug interaction risk.
- ➢ Filter out unsafe combinations and rank promising three-drug combinations for further study.
- ➢ Bring together biomedical NLP, molecular generation, feature engineering, and graph-based prediction in one project.

Main datasets and components

- ➢ Datasets: ChEMBL, DrugBank, SIDER, STITCH, and related supporting sources.
- ➢ Model ideas: BioBERT for text understanding, JT-VAE for molecule generation, and Pairwise / 3-Way GNNs for interaction prediction.

How I contributed within the team

- ★ Built the DrugBank preprocessing and feature engineering workflow.
- ★ Converted raw DrugBank XML into structured parquet data tables.
- ★ Engineered interaction, CYP enzyme, and safety-related features to support downstream modeling.
- ★ Helped communicate the pipeline clearly through presentations and GitHub organization.

Phase 1 – Project Proposal and Planning

Phase 1

This phase defined the problem, the architecture, the datasets, and the overall capstone direction.

Main goals of Phase 1

- Frame the project around a real problem: identifying safer and more effective multi-drug combinations.
- Decide on a large end-to-end pipeline rather than a single isolated model.
- Map out the major modules that would later become team responsibilities.

Key ideas proposed

- BioBERT would interpret mechanism text and produce embeddings.
- A molecular VAE / JT-VAE would generate chemically valid candidate molecules.
- DrugBank and related datasets would provide interaction and safety context.
- Pairwise and three-way GNN models would score candidate combinations.
- A ranking stage would return the safest, most promising outputs.

What I learned

- How to break a complex AI product into smaller modules.
- How data, modeling, and user-facing design fit together.
- Why feasibility, dataset access, and timeline matter in project planning.

Why Phase 1 mattered:

It set the roadmap for the entire semester and made the team roles clearer before implementation began.

Phase 2 – Data Pipeline and Exploratory Analysis

Phase 2

This was the phase where I made my biggest direct contribution through the DrugBank workflow.

My DrugBank data pipeline work

- Worked with a 1.76 GB DrugBank XML file in Google Colab.
- Used structured stream parsing to extract the data we actually needed instead of loading the entire XML into memory.
- Created four reusable parquet tables: drugs, drug_interactions, targets, and enzymes.
- Reduced the initial extraction from ~1 million duplicated rows to about 19,830 unique drugs.

Feature engineering and preprocessing

- Standardized data types and handled missing values.
- Engineered interaction count features such as subject interactions, affected-drug interactions, and total interaction degree.
- Built CYP enzyme and metabolic complexity features, including unique CYP counts and CYP-associated records.
- Created a safety complexity score to summarize interaction and metabolic burden in a more interpretable way.
- Prepared an active subset of 4,657 drugs and created train/validation/test splits.

EDA insights I explained

- Most drugs had relatively few interactions, while a small subset had many.
- Many drugs had little or no CYP information, showing data sparsity.
- The distributions were long-tailed, which is common in pharmacology because a few well-studied drugs dominate the data.

Also completed

I created and presented the Phase 2 presentation, making the DrugBank pipeline easier for the audience to follow.

Phase 3 – Model Architecture, Training, and Evaluation

This phase focused on the full AI system design and how each team component connected.

Major model components

- BioBERT: converts mechanism text into a 768-dimensional embedding.
- Embedding mapper: compresses the BioBERT output into the JT-VAE latent space.
- JT-VAE: generates chemically valid molecules from the mapped latent representation.
- 3-Way GNN: uses engineered drug and triplet features to predict interaction probability and support safety ranking.

How my DrugBank work supported Phase 3

- My engineered DrugBank features provided structured input for the interaction prediction stage.
- The project slides describe this as 19,830 processed drugs with enhanced features such as similarity, centrality, CYP overlap, and severity-related signals.
- This helped bridge raw biomedical data and the final GNN component.

What I did personally in this phase

- Reviewed the model architecture materials and notebooks to understand how the end-to-end system worked.
- Focused especially on how DrugBank data connected with the GNN and the larger pipeline.

Key learning

This phase helped me understand how data engineering supports downstream machine learning, even when the final predictive model is built by another teammate.

Phase 4 – Complete Pipeline Integration

Phase 4

The project moved from separate pieces toward one connected workflow.

1. User Input
Therapeutic mechanism in natural language

2. BioBERT
Encode text into a biomedical embedding

3. JT-VAE
Generate candidate molecules

4. DrugBank Features
Look up / engineer drug-level features

5. GNN Scoring
Evaluate pairwise and 3-way safety risk

6. Output
Return top-ranked safe combinations

What Phase 4 showed

- The project was not just a set of disconnected notebooks; it was designed as a workflow where each module passed information to the next.
- This made it easier to explain how user input could eventually become ranked drug-combination suggestions.

My perspective in this phase

- I reviewed the integration materials to better understand where my DrugBank features fit in the larger system.
- This also helped me communicate the project more clearly in GitHub and portfolio materials.

Phase 5 – Final Presentation and End-to-End Story

Phase 5

The last phase focused on communicating the complete system, the final pipeline, and the value of the project.

What the final deliverable highlighted

- ➢ The problem of harmful multi-drug interactions and why screening every 3-drug combination manually is unrealistic.
- ➢ The full end-to-end architecture: ingestion, data processing, feature extraction, model fit, and output.
- ➢ Why JT-VAE was chosen for molecule generation and how the user pipeline worked.
- ➢ How DrugBank feature engineering strengthened the interaction-prediction stage.

How I contributed to the final result

- ➢ My DrugBank work carried forward into the final narrative of the project.
- ➢ The final presentation explicitly described the DrugBank features I helped build, including molecular similarity, graph centrality, CYP enzyme information, and interaction severity context.
- ➢ I also helped by organizing the group GitHub repository so the project materials were cleaner and easier to submit.

Professional takeaway

- ➢ This phase showed me how technical work has to be communicated clearly, not just coded.
- ➢ A strong final presentation connects the data, the model, the workflow, and the real-world use case.

Key Skills I Built Across the Project

This phase summary is meant to show what I can contribute in future AI and data projects.

Technical skills

- Data preprocessing and cleaning in Python
- Large-file handling and structured XML parsing
- Feature engineering for biomedical / drug datasets
- Parquet-based data organization
- Exploratory Data Analysis and distribution interpretation
- Understanding how engineered features feed downstream ML models

Project and collaboration skills

- Working within a team-based capstone project
- Creating clear technical presentations
- Reviewing related notebooks to understand the full pipeline context
- Organizing GitHub repositories professionally
- Connecting individual tasks to a larger end-to-end AI system

Most relevant tools

- Python, Google Colab
- pandas, NumPy, scikit-learn
- lxml, Parquet workflows
- NetworkX and drug-interaction graph concepts
- RDKit concepts for molecular data
- GitHub for version control and portfolio presentation

CompoundIQ was a group project, but this portfolio summary clearly shows the areas I personally owned and the skills I developed throughout all five phases.